

Owned Requirements: FR-001–FR-008, SEC-001–SEC-005

Owned Components: Auth/User Module, Complaint Module

3.4 Component Descriptions

3.4.1 Auth/User Module

The Auth/User Module manages resident account creation, authentication, logout and role-based access control. It validates registration information, establishes authenticated sessions and ensures that users can access only the functions permitted for their assigned role. The module maintains user-related information and supports secure handling of authentication credentials. It is responsible for implementing FR-001 to FR-004 and supporting security requirements SEC-001 to SEC-004.

3.4.2 Complaint Module

The Complaint Module manages the creation and initial submission of complaints by authenticated residents. It validates mandatory complaint information including title, category, description and location, handles optional supporting attachments according to configured restrictions, and generates a unique Complaint ID after successful submission. The module implements FR-005 to FR-008 and supports SEC-003 and SEC-005 through input and attachment validation.

3.8 Traceability to Requirements

The following rows cover the requirements owned by Riya and map each requirement to its component and test cases.

Requirement ID	Component	Test Case IDs
FR-001	Auth/User Module	TC-AUTH-01, TC-AUTH-02
FR-002	Auth/User Module	TC-AUTH-03, TC-AUTH-04
FR-003	Auth/User Module	TC-AUTH-05, TC-AUTH-06
FR-004	Auth/User Module	TC-AUTH-07, TC-AUTH-08
FR-005	Complaint Module	TC-CMP-01, TC-CMP-02
FR-006	Complaint Module	TC-CMP-03, TC-CMP-04
FR-007	Complaint Module	TC-CMP-05, TC-CMP-06
FR-008	Complaint Module	TC-CMP-07, TC-CMP-08
SEC-001	Auth/User Module	TC-SEC-01, TC-SEC-02
SEC-002	Auth/User Module	TC-SEC-03, TC-SEC-04
SEC-003	Auth/User + Complaint Module	TC-SEC-05, TC-SEC-06
SEC-004	Auth/User Module	TC-SEC-07, TC-SEC-08
SEC-005	Complaint Module	TC-SEC-09, TC-SEC-10

3.9 Security Architecture – STRIDE

The Smart Complaint Management System applies role-based access control, secure credential handling, input validation, session protection and attachment restrictions to protect user and complaint information. Security controls are designed around the major threat categories represented by the STRIDE model.

STRIDE Threat	SCMS Threat	Security Control
Spoofing	Unauthorized user attempts to authenticate as another resident.	Authentication using registered credentials and secure password handling.
Tampering	User attempts to modify requests	Server-side input validation and

	or submit manipulated complaint data.	authorization checks.
Repudiation	User denies performing an authenticated operation.	Authenticated sessions and recording of complaint/status actions.
Information Disclosure	Unauthorized access to resident or complaint information.	Role-based access control and protected authenticated endpoints.
Denial of Service	Excessive or malformed requests affecting application availability.	Input validation, request restrictions and appropriate error handling.
Elevation of Privilege	Resident attempts to access staff/admin functionality.	Role-based authorization enforced on protected operations.

Security Requirement Mapping

- SEC-001: Passwords shall not be stored in plaintext.
- SEC-002: Residents should only access resident features, not staff or administrator features.
- SEC-003: The system should check and clean user input before using it.
- SEC-004: Users with invalid or expired login sessions should not be allowed to access protected pages.
- SEC-005: Complaint attachments shall be restricted by configured file type and size.

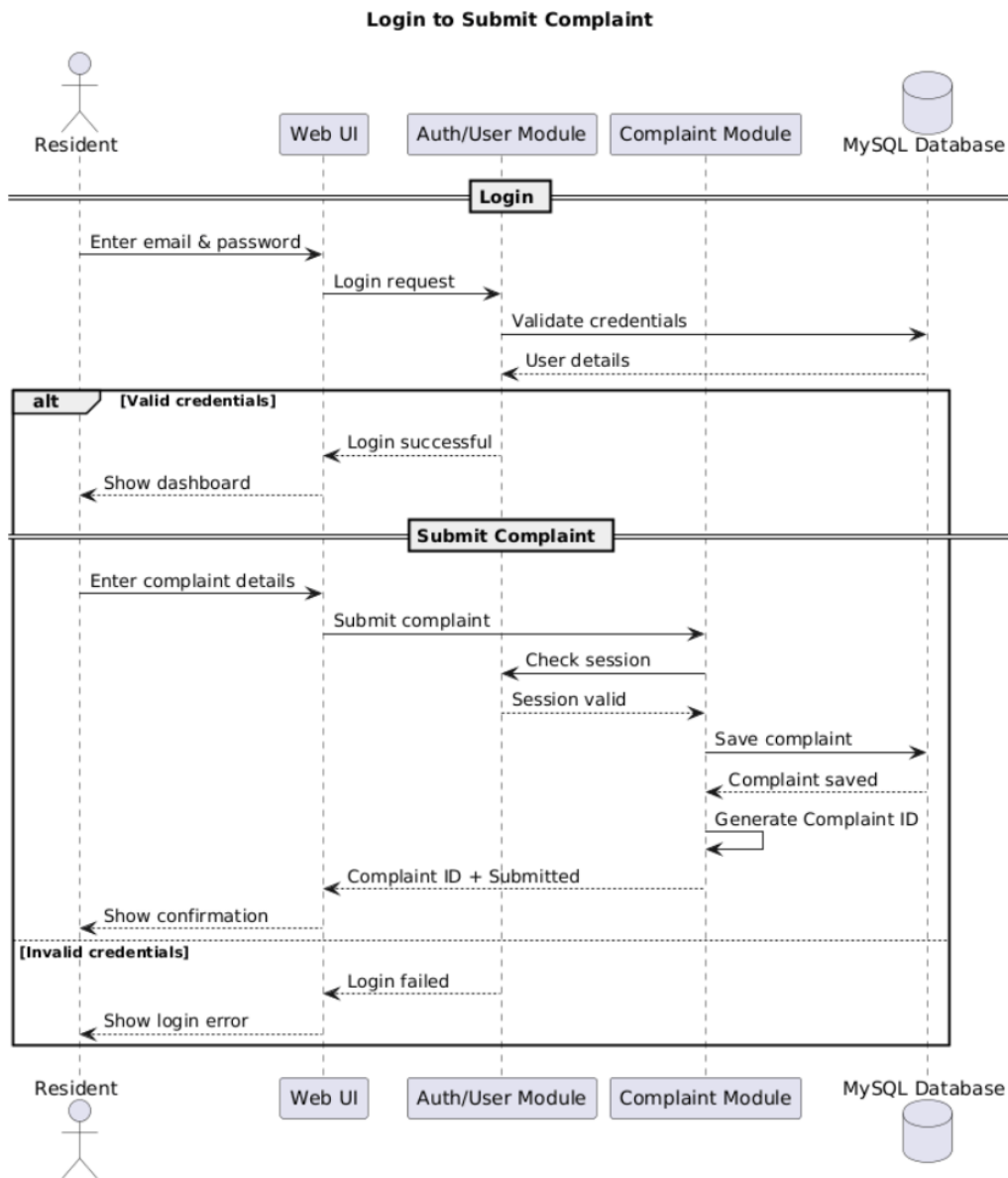
Security Boundaries

The main security boundary is established between the resident, authentication/authorization layer, application API, business modules and database:

Resident → Authentication / Authorization → Application API → Business Modules → MySQL Database

4.2 UML Sequence Diagram

4.2.1 Login → Submit Complaint



4.3 API Design

The following are proposed design-level API interfaces for the Auth/User and Complaint components. They define the expected method, request, response and error behavior for implementation.

Auth/User Module

Endpoint	Method	Purpose
/api/auth/register	POST	Create resident account
/api/auth/login	POST	Authenticate user
/api/auth/logout	POST	End authenticated session
/api/users/me	GET	Retrieve authenticated user's profile

POST /api/auth/register

Request: name, email, password, contact_number, unit_address

Response: Account-created message and generated user_id.

Errors: 400 – missing/invalid required information; 409 – email/account already exists.

POST /api/auth/login

Request: email, password

Response: Login-success message and authenticated session.

Errors: 400 – invalid request; 401 – invalid credentials.

POST /api/auth/logout

Request: authenticated session/token

Response: Logout-success message.

Errors: 401 – invalid or expired session.

GET /api/users/me

Request: Authenticated session/token.

Response: User profile containing user_id, name, email, contact_number, unit_address and role.

Errors: 401 – invalid or expired session.

Complaint Module

Endpoint	Method	Purpose
/api/complaints	POST	Submit a new complaint
/api/complaints/{complaint_id}	GET	Retrieve complaint details
/api/complaints/{complaint_id}/attachments	POST	Upload supporting attachment

POST /api/complaints

Request: title, category_id, description, location

Response: successful submission message, unique complaint_id and status=Submitted.

Errors: 400 – missing/invalid complaint information; 401 – user not authenticated; 422 – invalid input.

POST /api/complaints/{complaint_id}/attachments

Request: multipart file upload

Response: attachment-uploaded message.

Errors: 400 – invalid file; 401 – user not authenticated; 413 – file exceeds configured size; 415 – unsupported file type.

GET /api/complaints/{complaint_id}

Request: complaint_id path parameter and authenticated session/token.

Response: Complaint details including complaint_id, title, category, description, location, status, created_at and attachment reference if applicable.

Errors: 401 – user not authenticated; 403 – user not permitted to access the complaint; 404 – complaint not found.